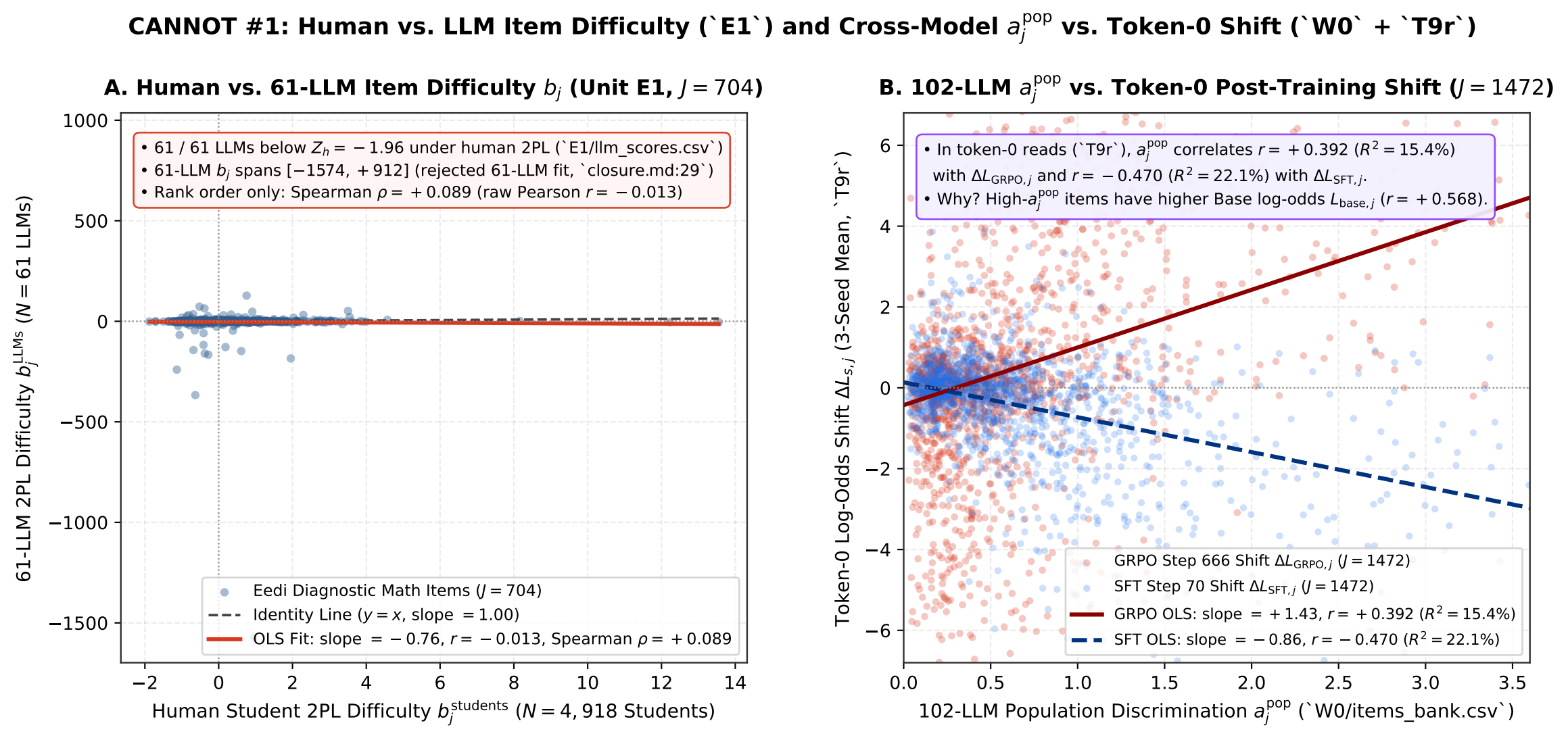

CANNOT #1: Human vs. LLM Item Difficulty (`E1`) and Cross-Model a_j^{pop} vs. Token-0 Shift (`W0` + `T9r`)
A. Human vs. 61-LLM Item Difficulty b_j (Unit E1, $J = 704$)
• 61 / 61 LLMs below $Z_h = -1.96$ under human 2PL (`E1/llm_scores.csv`)
• 61-LLM b_j spans $[-1574, +912]$ (rejected 61-LLM fit, `closure.md:29`)
• Rank order only: Spearman $\rho = +0.089$ (raw Pearson $r = -0.013$)
Eedi Diagnostic Math Items ($J = 704$)
Identity Line ($y = x$, slope $= 1.00$)
OLS Fit: slope $= -0.76$, $r = -0.013$, Spearman $\rho = +0.089$
61-LLM 2PL Difficulty b_j^{LLMs} ($N = 61$ LLMs)
Human Student 2PL Difficulty $b_j^{students}$ ($N = 4,918$ Students)
B. 102-LLM a_j^{pop} vs. Token-0 Post-Training Shift ($J = 1472$)
• In token-0 reads (`T9r`), a_j^{pop} correlates $r = +0.392$ ($R^2 = 15.4\%$) with $\Delta L_{GRPO,j}$ and $r = -0.470$ ($R^2 = 22.1\%$) with $\Delta L_{SFT,j}$.
• Why? High-a_j^{pop} items have higher Base log-odds $L_{base,j}$ ($r = +0.568$).
GRPO Step 666 Shift $\Delta L_{GRPO,j}$ ($J = 1472$)
SFT Step 70 Shift $\Delta L_{SFT,j}$ ($J = 1472$)
GRPO OLS: slope $= +1.43$, $r = +0.392$ ($R^2 = 15.4\%$)
SFT OLS: slope $= -0.86$, $r = -0.470$ ($R^2 = 22.1\%$)
Token-0 Log-Odds Shift $\Delta L_{s,j}$ (3-Seed Mean, `T9r`)
102-LLM Population Discrimination a_j^{pop} (`W0/items_bank.csv`)